

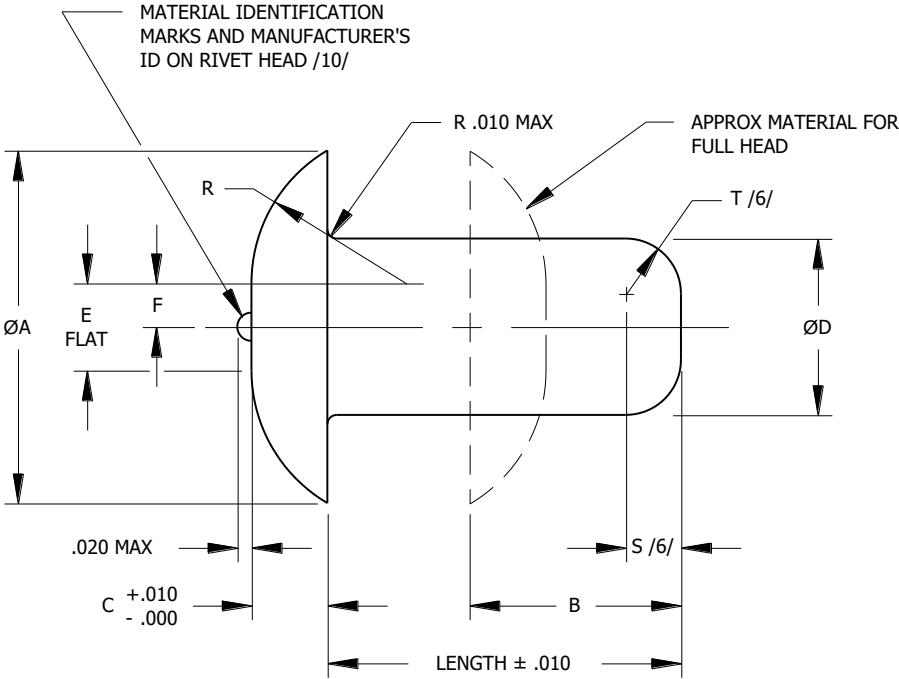


NATIONAL AEROSPACE STANDARD

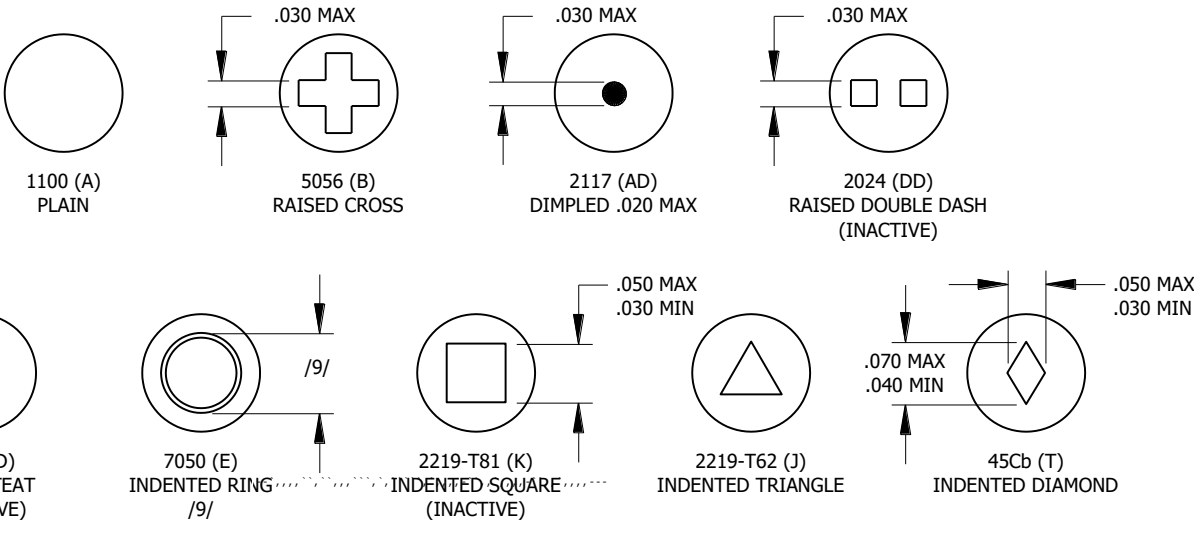
© COPYRIGHT 2025 AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA, INC. ALL RIGHTS RESERVED



FED SUPPLY CLASS
5320



MATERIAL IDENTIFICATION MARKS



THE INITIAL RELEASE OF THIS DOCUMENT SUPERSEDES MS20470, REVISION H, PART NUMBERS REMAIN MS20470

THIRD ANGLE PROJECTION	CUSTODIAN NATIONAL AEROSPACE STANDARDS COMMITTEE	REVISION 3
PROCUREMENT SPECIFICATION NASM5674	TITLE RIVET, SOLID, UNIVERSAL HEAD, ALUMINUM ALLOY AND TITANIUM COLUMBIUM ALLOY	CLASSIFICATION PART STANDARD NASM20470 SHEET 1 OF 9

USE OF OR RELIANCE UPON THIS DOCUMENT OR ANY NATIONAL AEROSPACE STANDARD IS ENTIRELY VOLUNTARY. AIA DOES NOT QUALIFY SUPPLIERS OR CERTIFY CONFORMANCE OF ITEMS PRODUCED UNDER NATIONAL AEROSPACE STANDARDS. AIA MAKES NO REPRESENTATION OR CLAIM RESPECTING (1) THE SUITABILITY OF ITEMS FOR ANY PARTICULAR APPLICATION OR (2) THE ACCURACY OF THE INFORMATION CONTAINED HEREIN. AIA IS NOT RESPONSIBLE FOR ANY ERRORS OR OMISSIONS. AIA IS NOT RESPONSIBLE FOR ANY REPRODUCTION OR DISTRIBUTION OF THIS DOCUMENT WITHOUT THE WRITTEN PERMISSION OF AIA. AIA IS NOT RESPONSIBLE FOR ANY REPRODUCTION OR DISTRIBUTION OF THIS DOCUMENT WITHOUT THE WRITTEN PERMISSION OF AIA. AIA IS NOT RESPONSIBLE FOR ANY REPRODUCTION OR DISTRIBUTION OF THIS DOCUMENT WITHOUT THE WRITTEN PERMISSION OF AIA.

AIRBUS | NormMaster
AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA, INC.
1000 WILSON BLVD.
ARLINGTON, VA 22209
THIS DRAWING SUPERSEDES ALL ANTECEDENT STANDARD DRAWINGS FOR THE SAME
PRODUCT AND SHALL BECOME EFFECTIVE NO LATER THAN SIX MONTHS FROM THE LAST
REVISION DATE.
FORM 21-01

REVISION DATE: JANUARY 31, 2025
ISSUE DATE: AUGUST 1998

TABLE I – RIVET DIMENSIONS

NOM DIA		ØD	ØA	B	C	E	F	R	S	T RAD
1/16	.062	+.003 - .001 /15/	.125 ± .006	.094	.027	.031	.015	.054	.016	.019
3/32	.094		.187 ± .009	.141	.040	.046	.023	.082	.023	.029
1/8	.125		.250 ± .012	.188	.054	.062	.031	.108	.031	.039
5/32	.156		.312 ± .016	.234	.067	.078	.039	.135	.039	.049
3/16	.187		.375 ± .019	.281	.080	.093	.046	.164	.047	.059
7/32	.219		.437 ± .022	.328	.093	.109	.054	.193	.054	.069
1/4	.250		.500 ± .025	.375	.107	.125	.062	.217	.062	.078
5/16	.312		.625 ± .031	.469	.133	.156	.078	.272	.078	.098
3/8	.375		.750 ± .037	.562	.161	.187	.093	.328	.094	.117

MATERIAL:

ALUMINUM 1100-F (UNS A91100) AND ALUMINUM ALLOYS 2017-T4 (UNS A92017), 2024-T4 (UNS A92024), 2117-T4 (UNS A92117), 2219-T81 (UNS A92219), 2219-T62 (UNS A92219), 5056-H32 (UNS A95056), 7050-T73 (UNS A97050) PER QQ-A-430.
TITANIUM COLUMBIUM ALLOY 45Cb (UNS R58450) PER AMS4982.

FINISH:

FOR 2024 AL:
ANODIZE PER MIL-PRF-8625, TYPE II, CLASS 1, CLEAR (A CODE), OR
ANODIZE PER MIL-PRF-8625, TYPE II, CLASS 1, DICHROMATE SEAL (D CODE), OR
ANODIZE PER MIL-PRF-8625, TYPE II, CLASS 2 WITH SPECIFIED COLOR (N CODE), OR
ANODIZE PER MIL-PRF-8625, TYPE II, CLASS 1, DICHROMATE SEAL IS PROHIBITED (G CODE), OR
ANODIZE PER MIL-PRF-8625, TYPE II, CLASS 2 WITH SPECIFIED COLOR, DICHROMATE SEAL IS
PROHIBITED (P CODE).

FOR 1100 AL:
CHEMICAL SURFACE TREATMENT PER MIL-DTL-5541, TYPE I, CLASS 1A, GOLD/YELLOW (NO CODE), OR
CHEMICAL SURFACE TREATMENT PER MIL-DTL-5541, TYPE II, CLASS 1A (C CODE).

FOR 2017, 2117, 2219, 5056 AND 7050 AL:
 CHEMICAL SURFACE TREATMENT PER MIL-DTL-5541, TYPE I, GOLD/YELLOW (NO CODE), OR
 CHEMICAL SURFACE TREATMENT PER MIL-DTL-5541, TYPE I, CLASS 1A, CLEAR (S CODE)
 (CLEAR FILM SHALL BE USED ONLY WHEN AUTHORIZED), OR
 CHEMICAL SURFACE TREATMENT PER MIL-DTL-5541, TYPE II, CLASS 1A (C CODE), OR
 ANODIZE PER MIL-PRF-8625, TYPE II, CLASS 1, CLEAR (A CODE), OR
 ANODIZE PER MIL-PRF-8625, TYPE II, CLASS 1, DICHROMATE SEAL (D CODE), OR
 ANODIZE PER MIL-PRF-8625, TYPE II, CLASS 2 WITH SPECIFIED COLOR (N CODE), OR
 ANODIZE PER MIL-PRF-8625, TYPE II, CLASS 1, DICHROMATE SEAL IS PROHIBITED (G CODE), OR
 ANODIZE PER MIL-PRF-8625, TYPE II, CLASS 2 WITH SPECIFIED COLOR, DICHROMATE SEAL IS
 PROHIBITED (P CODE).

FOR TITANIUM COLUMBIUM, NO FINISH REQUIRED.

REVISION

3

NASM20470

SHEET 2



NATIONAL AEROSPACE STANDARD

© COPYRIGHT 2025 AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA, INC. ALL RIGHTS RESERVED



CODE:

3

MS20470 AD 4-12 A -5

INDICATES ADDITIONAL .031 (1/32 INCH) INCREMENT IN LENGTH

FINISH CODE:

"NO CODE" INDICATES CHEMICAL SURFACE TREATMENT PER MIL-DTL-5541, CLASS 1A, GOLD/YELLOW

"C" CODE: CHEMICAL SURFACE TREATMENT PER MIL-DTL-5541, TYPE II, CLASS 1A

"A" CODE: ANODIZE PER MIL-PRF-8625, TYPE II, CLASS 1, CLEAR

"D" CODE: ANODIZE PER MIL-PRF-8625, TYPE II, CLASS 1, DICHROMATE SEAL

"G" CODE: ANODIZE PER MIL-PRF-8625, TYPE II, CLASS 1, DICHROMATE SEAL PROHIBITED

"F" CODE: OPTION OF ANY FINISH EXCEPT AS NOTED

"N" CODE: ANODIZE PER MIL-PRF-8625, TYPE II, CLASS 2

"P" CODE: ANODIZE PER MIL-PRF-8625, TYPE II, CLASS 2, DICHROMATE SEAL PROHIBITED

"S" CODE: CHEMICAL SURFACE TREATMENT PER MIL-DTL-5541, CLASS 1A, CLEAR

INDICATES LENGTH IN .063 (1/16 INCH) OF AN INCH

INDICATES DIAMETER IN .031 (1/32 INCH) OF AN INCH

MATERIAL CODE:

"A" CODE: 1100-F ALUMINUM

"AD" CODE: 2117-T4 ALUMINUM ALLOY

"B" CODE: 5056-H32 ALUMINUM ALLOY

"D" CODE: 2017-T4 ALUMINUM ALLOY – INACTIVE FOR NEW DESIGN

"DD" CODE: 2024-T4 ALUMINUM ALLOY – INACTIVE FOR NEW DESIGN

"E" CODE: 7050-T73 ALUMINUM ALLOY

"J" CODE: 2219-T62 ALUMINUM ALLOY

"K" CODE: 2219-T81 ALUMINUM ALLOY – INACTIVE FOR NEW DESIGN

"T" CODE: TITANIUM COLUMBIUM ALLOY 45Cb

BASIC PART NUMBER

EXAMPLES OF PART NUMBER:

3

MS20470AD4-12G-5 = ALUMINUM ALLOY 2117-T4 RIVET, Ø.125 (Ø 1/8 INCH) (.750 + .031 (3/4 INCH + 1/32 INCH)) LONG, ANODIZE CLEAR, DICHROMATE SEAL PROHIBITED

MS20470E4-12N-5 = ALUMINUM ALLOY 7050-T73 RIVET, Ø.125 (Ø 1/8 INCH) (.750 + .031 (3/4 INCH + 1/32 INCH)) LONG, ANODIZE PER MIL-PRF-8625, TYPE II, CLASS 2

REVISION

3

NASM20470

SHEET 3



NATIONAL AEROSPACE STANDARD

© COPYRIGHT 2025 AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA, INC. ALL RIGHTS RESERVED



NOTES:

- (1) DIMENSIONS ARE IN INCHES AND APPLY AFTER FINISH UNLESS OTHERWISE SPECIFIED.
- (2) INTERCHANGEABILITY RELATION WITH AN470 RIVETS: MS20470 RIVETS OF LIKE DASH NUMBERS ARE UNIVERSALLY, FUNCTIONALLY, AND DIMENSIONALLY INTERCHANGEABLE.
- (3) THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN.
- (4) UNLESS OTHERWISE SPECIFIED HEREIN, REFERENCED DOCUMENTS SHALL BE THE ISSUE IN EFFECT ON DATE OF MANUFACTURE. HOWEVER, EXISTING MATERIAL INVENTORY CERTIFIED TO A PREVIOUS REVISION OF THE APPLICABLE MATERIAL SPECIFICATION(S) IS ACCEPTABLE FOR USE UNTIL DEPLETION.
- (5) TOLERANCES UNLESS OTHERWISE SPECIFIED: DECIMALS: ± .010.
- /6/ RIVETS MAY BE FURNISHED WITH PLAIN ENDS, OR ENDS CHAMFERED UP TO 20° TO THE "S" DIMENSION, OR MAY HAVE A RADIUS TO THE "T" DIMENSION.
- ③ (7) FOR PART NUMBERS OF RIVET LENGTHS NOT LISTED, .063 (1/16 INCH) LENGTH INCREMENTS REQUIRED.
- (8) ADDITIONAL 1/32 LENGTH INCREMENTS MAY BE OBTAINED BY ADDING -5 AFTER THE LAST DIGIT OR LETTER IN PART NUMBER.
- /9/ OUTER DIAMETER OF INDENTED RING (7050 RIVETS) SHALL BE A MAXIMUM OF 75% OF RIVET SHANK DIAMETER. THE MAXIMUM DEPTH OF THE RING SHALL BE .010 INCH. INDENTED RING IS REQUIRED ON 7050 RIVETS MANUFACTURED AFTER 2 JULY 1990.
- /10/ MANUFACTURER'S IDENTIFICATION IS REQUIRED TO BE DEPRESSED ON RIVET HEADS WITH A SHANK DIAMETER .125 INCH AND LARGER MANUFACTURED AFTER 2 JULY 1990.
- /11/ ALUMINUM ALLOY 2017-T4 RIVETS WITH 33 KSI MINIMUM UNDRIVEN SHEAR STRENGTHS ARE ACCEPTABLE UNTIL 2 JULY 1990.
- (12) REMOVE ALL BURRS AND SHARP EDGES.
- ③ (13) DIMENSIONING AND TOLERANCING PER ASME Y14.5-1982.
- (14) UNLESS OTHERWISE SPECIFIED, PART INVENTORY MANUFACTURED TO PREVIOUS REVISIONS OF THE APPLICABLE DRAWING OR SPECIFICATION MAY BE PROCURED AND USED UNTIL STOCK IS DEPLETED.
- /15/ TOLERANCE CHANGE APPLIES TO RIVETS MANUFACTURED AFTER 2 JULY 1990.

REVISION
3

NASM20470
SHEET 4

USE OF OR RELIANCE UPON THIS DOCUMENT OR ANY NATIONAL AEROSPACE STANDARD IS ENTIRELY VOLUNTARY. AIA DOES NOT QUALIFY SUPPLIERS OR CERTIFY CONFORMANCE OF ITEMS PRODUCED UNDER NATIONAL AEROSPACE STANDARDS. AIA MAKES NO REPRESENTATION OR CLAIM RESPECTING (1) THE SUITABILITY OF ITEMS FOR ANY PARTICULAR APPLICATION OR (2) THE FITNESS OF ANY ITEM FOR ANY PARTICULAR PURPOSE. THIS DOCUMENT IS THE PROPERTY OF AIA AND IS LOANED TO YOUR ORGANIZATION. IT IS TO BE RETURNED TO AIA UPON REQUEST. NO REPRODUCTION OR NETWORKING PERMITTED WITHOUT LICENSE FROM ACCURIS

AIRBUS | NormMaster
THIS DRAWING SUPERSEDES ALL ANTECEDENT STANDARD DRAWINGS FOR THE SAME PRODUCT AND SHALL BECOME EFFECTIVE NO LATER THAN SIX MONTHS FROM THE LAST REVISION DATE.
FORM 21-01



NATIONAL AEROSPACE STANDARD

© COPYRIGHT 2025 AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA, INC. ALL RIGHTS RESERVED



TABLE II – DASH NUMBER LENGTH DIMENSIONS

ALUMINUM 1100-F	ALUMINUM 2117-T4	ALUMINUM 5056-H32	ALUMINUM 2017-T4 INACTIVE	ALUMINUM 2024-T4 INACTIVE	ALUMINUM 7050-T73	ALUMINUM 2219-T81 INACTIVE	TITANIUM 45Cb	ALUMINUM 2219-T62	LENGTH
DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	
Ø1/16 NOM									
A2-3	AD2-3	B2-3	D2-3	DD2-3	E2-3	K2-3	T2-3	J2-3	.188
A2-4	AD2-4	B2-4	D2-4	DD2-4	E2-4	K2-4	T2-4	J2-4	.250
A2-5	AD2-5	B2-5	D2-5	DD2-5	E2-5	K2-5	T2-5	J2-5	.312
A2-6	AD2-6	B2-6	D2-6	DD2-6	E2-6	K2-6	T2-6	J2-6	.375
A2-7	AD2-7	B2-7	D2-7	DD2-7	E2-7	K2-7	T2-7	J2-7	.438
A2-8	AD2-8	B2-8	D2-8	DD2-8	E2-8	K2-8	T2-8	J2-8	.500
A2-9	AD2-9	B2-9	D2-9	DD2-9	E2-9	K2-9	T2-9	J2-9	.562
A2-10	AD2-10	B2-10	D2-10	DD2-10	E2-10	K2-10	T2-10	J2-10	.625
A2-12	AD2-12	B2-12	D2-12	DD2-12	E2-12	K2-12	T2-12	J2-12	.750
A2-14	AD2-14	B2-14	D2-14	DD2-14	E2-14	K2-14	T2-14	J2-14	.875
A2-16	AD2-16	B2-16	D2-16	DD2-16	E2-16	K2-16	T2-16	J2-16	1.000
A2-18	AD2-18	B2-18	D2-18	DD2-18	E2-18	K2-18	T2-18	J2-18	1.125
A2-20	AD2-20	B2-20	D2-20	DD2-20	E2-20	K2-20	T2-20	J2-20	1.250

ALUMINUM 1100-F	ALUMINUM 2117-T4	ALUMINUM 5056-H32	ALUMINUM 2017-T4 INACTIVE	ALUMINUM 2024-T4 INACTIVE	ALUMINUM 7050-T73	ALUMINUM 2219-T81 INACTIVE	TITANIUM 45Cb	ALUMINUM 2219-T62	LENGTH
DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	
Ø3/32 NOM									
A3-3	AD3-3	B3-3	D3-3	DD3-3	E3-3	K3-3	T3-3	J3-3	.188
A3-4	AD3-4	B3-4	D3-4	DD3-4	E3-4	K3-4	T3-4	J3-4	.250
A3-5	AD3-5	B3-5	D3-5	DD3-5	E3-5	K3-5	T3-5	J3-5	.312
A3-6	AD3-6	B3-6	D3-6	DD3-6	E3-6	K3-6	T3-6	J3-6	.375
A3-7	AD3-7	B3-7	D3-7	DD3-7	E3-7	K3-7	T3-7	J3-7	.438
A3-8	AD3-8	B3-8	D3-8	DD3-8	E3-8	K3-8	T3-8	J3-8	.500
A3-9	AD3-9	B3-9	D3-9	DD3-9	E3-9	K3-9	T3-9	J3-9	.562
A3-10	AD3-10	B3-10	D3-10	DD3-10	E3-10	K3-10	T3-10	J3-10	.625
A3-12	AD3-12	B3-12	D3-12	DD3-12	E3-12	K3-12	T3-12	J3-12	.750
A3-14	AD3-14	B3-14	D3-14	DD3-14	E3-14	K3-14	T3-14	J3-14	.875
A3-16	AD3-16	B3-16	D3-16	DD3-16	E3-16	K3-16	T3-16	J3-16	1.000
A3-18	AD3-18	B3-18	D3-18	DD3-18	E3-18	K3-18	T3-18	J3-18	1.125
A3-20	AD3-20	B3-20	D3-20	DD3-20	E3-20	K3-20	T3-20	J3-20	1.250
A3-22	AD3-22	B3-22	D3-22	DD3-22	E3-22	K3-22	T3-22	J3-22	1.375

REVISION
3

NASM20470
SHEET 5

USE OF OR RELIANCE UPON THIS DOCUMENT OR ANY NATIONAL AEROSPACE STANDARD IS ENTIRELY VOLUNTARY. AIA DOES NOT QUALIFY SUPPLIERS OR CERTIFY CONFORMANCE OF ITEMS PRODUCED UNDER NATIONAL AEROSPACE STANDARDS. AIA MAKES NO REPRESENTATION OR CLAIM RESPECTING (1) THE SUITABILITY OF ITEMS FOR ANY PARTICULAR APPLICATION OR (2) THE ACCURACY OF THE INFORMATION CONTAINED HEREIN. AIA IS NOT RESPONSIBLE FOR ANY ERRORS OR OMISSIONS. AIA IS NOT RESPONSIBLE FOR ANY REPRODUCTION OR DISTRIBUTION OF THIS DOCUMENT WITHOUT THE WRITTEN PERMISSION OF AIA. AIA IS NOT RESPONSIBLE FOR ANY REPRODUCTION OR DISTRIBUTION OF THIS DOCUMENT WITHOUT THE WRITTEN PERMISSION OF AIA. AIA IS NOT RESPONSIBLE FOR ANY REPRODUCTION OR DISTRIBUTION OF THIS DOCUMENT WITHOUT THE WRITTEN PERMISSION OF AIA.



NATIONAL AEROSPACE STANDARD

© COPYRIGHT 2025 AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA, INC. ALL RIGHTS RESERVED



TABLE II – DASH NUMBER LENGTH DIMENSIONS (CONTINUED)

ALUMINUM 1100-F	ALUMINUM 2117-T4	ALUMINUM 5056-H32	ALUMINUM 2017-T4 INACTIVE	ALUMINUM 2024-T4 INACTIVE	ALUMINUM 7050-T73	ALUMINUM 2219-T81 INACTIVE	TITANIUM 45Cb	ALUMINUM 2219-T62	LENGTH
DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	
Ø1/8 NOM									
A4-3	AD4-3	B4-3	D4-3	DD4-3	E4-3	K4-3	T4-3	J4-3	.188
	AD4-3-5	B4-3-5				K4-3-5	T4-3-5	J4-3-5	.218
A4-4	AD4-4	B4-4	D4-4	DD4-4	E4-4	K4-4	T4-4	J4-4	.250
	AD4-4-5	B4-4-5				K4-4-5	T4-4-5	J4-4-5	.281
A4-5	AD4-5	B4-5	D4-5	DD4-5	E4-5	K4-5	T4-5	J4-5	.312
A4-6	AD4-6	B4-6	D4-6	DD4-6	E4-6	K4-6	T4-6	J4-6	.375
A4-7	AD4-7	B4-7	D4-7	DD4-7	E4-7	K4-7	T4-7	J4-7	.438
A4-8	AD4-8	B4-8	D4-8	DD4-8	E4-8	K4-8	T4-8	J4-8	.500
A4-9	AD4-9	B4-9	D4-9	DD4-9	E4-9	K4-9	T4-9	J4-9	.562
A4-10	AD4-10	B4-10	D4-10	DD4-10	E4-10	K4-10	T4-10	J4-10	.625
	AD4-11	B4-11				K4-11	T4-11	J4-11	.687
A4-12	AD4-12	B4-12	D4-12	DD4-12	E4-12	K4-12	T4-12	J4-12	.750
A4-14	AD4-14	B4-14	D4-14	DD4-14	E4-14	K4-14	T4-14	J4-14	.875
A4-16	AD4-16	B4-16	D4-16	DD4-16	E4-16	K4-16	T4-16	J4-16	1.000
A4-18	AD4-18	B4-18	D4-18	DD4-18	E4-18	K4-18	T4-18	J4-18	1.125
A4-20	AD4-20	B4-20	D4-20	DD4-20	E4-20	K4-20	T4-20	J4-20	1.250
A4-22	AD4-22	B4-22	D4-22	DD4-22	E4-22	K4-22	T4-22	J4-22	1.375
A4-24	AD4-24	B4-24	D4-24	DD4-24	E4-24	K4-24	T4-24	J4-24	1.500

ALUMINUM 1100-F	ALUMINUM 2117-T4	ALUMINUM 5056-H32	ALUMINUM 2017-T4 INACTIVE	ALUMINUM 2024-T4 INACTIVE	ALUMINUM 7050-T73	ALUMINUM 2219-T81 INACTIVE	TITANIUM 45Cb	ALUMINUM 2219-T62	LENGTH
DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	
Ø5/32 NOM									
A5-4	AD5-4	B5-4	D5-4	DD5-4	E5-4	K5-4	T5-4	J5-4	.250
A5-5	AD5-5	B5-5	D5-5	DD5-5	E5-5	K5-5	T5-5	J5-5	.312
A5-6	AD5-6	B5-6	D5-6	DD5-6	E5-6	K5-6	T5-6	J5-6	.375
A5-7	AD5-7	B5-7	D5-7	DD5-7	E5-7	K5-7	T5-7	J5-7	.438
A5-8	AD5-8	B5-8	D5-8	DD5-8	E5-8	K5-8	T5-8	J5-8	.500
A5-9	AD5-9	B5-9	D5-9	DD5-9	E5-9	K5-9	T5-9	J5-9	.562
A5-10	AD5-10	B5-10	D5-10	DD5-10	E5-10	K5-10	T5-10	J5-10	.625
	AD5-11					K5-11		J5-11	.687
A5-12	AD5-12	B5-12	D5-12	DD5-12	E5-12	K5-12	T5-12	J5-12	.750
A5-14	AD5-14	B5-14	D5-14	DD5-14	E5-14	K5-14	T5-14	J5-14	.875
A5-16	AD5-16	B5-16	D5-16	DD5-16	E5-16	K5-16	T5-16	J5-16	1.000
	AD5-17					K5-17			1.062
A5-18	AD5-18	B5-18	D5-18	DD5-18	E5-18	K5-18	T5-18	J5-18	1.125
	AD5-19					K5-19		J5-19	1.187
A5-20	AD5-20	B5-20	D5-20	DD5-20	E5-20	K5-20	T5-20	J5-20	1.250
A5-22	AD5-22	B5-22	D5-22	DD5-22	E5-22	K5-22	T5-22	J5-22	1.375
A5-24	AD5-24	B5-24	D5-24	DD5-24	E5-24	K5-24	T5-24	J5-24	1.500
A5-28	AD5-28	B5-28	D5-28	DD5-28	E5-28	K5-28	T5-28	J5-28	1.750

REVISION

3

NASM20470

SHEET 6

USE OF OR RELIANCE UPON THIS DOCUMENT OR ANY NATIONAL AEROSPACE STANDARD IS ENTIRELY VOLUNTARY. AIA DOES NOT QUALIFY SUPPLIERS OR CERTIFY CONFORMANCE OF ITEMS PRODUCED UNDER NATIONAL AEROSPACE STANDARDS. AIA MAKES NO REPRESENTATION OR CLAIM RESPECTING (1) THE SUITABILITY OF ITEMS FOR ANY PARTICULAR APPLICATION OR (2) THE ACCURACY OF THE INFORMATION CONTAINED HEREIN. THIS DOCUMENT IS THE PROPERTY OF AIA AND IS LOANED TO YOU. IT IS TO BE RETURNED TO AIA UPON REQUEST. NO REPRODUCTION OR NETWORKING PERMITTED WITHOUT LICENSE FROM ACCURIS.



NATIONAL AEROSPACE STANDARD

© COPYRIGHT 2025 AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA, INC. ALL RIGHTS RESERVED



TABLE II – DASH NUMBER LENGTH DIMENSIONS (CONTINUED)

ALUMINUM 1100-F	ALUMINUM 2117-T4	ALUMINUM 5056-H32	ALUMINUM 2017-T4 INACTIVE	ALUMINUM 2024-T4 INACTIVE	ALUMINUM 7050-T73	ALUMINUM 2219-T81 INACTIVE	TITANIUM 45Cb	ALUMINUM 2219-T62	LENGTH
DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	
Ø3/16 NOM									
	AD6-3								.187
A6-4	AD6-4	B6-4	D6-4	DD6-4	E6-4	K6-4	T6-4	J6-4	.250
A6-5	AD6-5	B6-5	D6-5	DD6-5	E6-5	K6-5	T6-5	J6-5	.312
A6-6	AD6-6	B6-6	D6-6	DD6-6	E6-6	K6-6	T6-6	J6-6	.375
A6-7	AD6-7	B6-7	D6-7	DD6-7	E6-7	K6-7	T6-7	J6-7	.438
A6-8	AD6-8	B6-8	D6-8	DD6-8	E6-8	K6-8	T6-8	J6-8	.500
A6-9	AD6-9	B6-9	D6-9	DD6-9	E6-9	K6-9	T6-9	J6-9	.562
A6-10	AD6-10	B6-10	D6-10	DD6-10	E6-10	K6-10	T6-10	J6-10	.625
	AD6-11				E6-11				.687
A6-12	AD6-12	B6-12	D6-12	DD6-12	E6-12	K6-12	T6-12	J6-12	.750
	AD6-13								.812
A6-14	AD6-14	B6-14	D6-14	DD6-14	E6-14	K6-14	T6-14	J6-14	.875
A6-16	AD6-16	B6-16	D6-16	DD6-16	E6-16	K6-16	T6-16	J6-16	1.000
A6-18	AD6-18	B6-18	D6-18	DD6-18	E6-18	K6-18	T6-18	J6-18	1.125
A6-20	AD6-20	B6-20	D6-20	DD6-20	E6-20	K6-20	T6-20	J6-20	1.250
A6-22	AD6-22	B6-22	D6-22	DD6-22	E6-22	K6-22	T6-22	J6-22	1.375
A6-24	AD6-24	B6-24	D6-24	DD6-24	E6-24	K6-24	T6-24	J6-24	1.500
A6-28	AD6-28	B6-28	D6-28	DD6-28	E6-28	K6-28	T6-28	J6-28	1.750
A6-32	AD6-32	B6-32	D6-32	DD6-32	E6-32	K6-32	T6-32	J6-32	2.000
	AD6-36								2.250
A6-40	AD6-40	B6-40	D6-40	DD6-40	E6-40	K6-40	T6-40	J6-40	2.500
A6-48	AD6-48	B6-48	D6-48	DD6-48	E6-48	K6-48	T6-48	J6-48	3.000

ALUMINUM 1100-F	ALUMINUM 2117-T4	ALUMINUM 5056-H32	ALUMINUM 2017-T4 INACTIVE	ALUMINUM 2024-T4 INACTIVE	ALUMINUM 7050-T73	ALUMINUM 2219-T81 INACTIVE	TITANIUM 45Cb	ALUMINUM 2219-T62	LENGTH
DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	
Ø7/32 NOM									
A7-6	AD7-6	B7-6	D7-6	DD7-6	E7-6	K7-6	T7-6	J7-6	.375
A7-7	AD7-7	B7-7	D7-7	DD7-7	E7-7	K7-7	T7-7	J7-7	.438
A7-8	AD7-8	B7-8	D7-8	DD7-8	E7-8	K7-8	T7-8	J7-8	.500
A7-9	AD7-9	B7-9	D7-9	DD7-9	E7-9	K7-9	T7-9	J7-9	.562
A7-10	AD7-10	B7-10	D7-10	DD7-10	E7-10	K7-10	T7-10	J7-10	.625
A7-12	AD7-12	B7-12	D7-12	DD7-12	E7-12	K7-12	T7-12	J7-12	.750
A7-14	AD7-14	B7-14	D7-14	DD7-14	E7-14	K7-14	T7-14	J7-14	.875
A7-16	AD7-16	B7-16	D7-16	DD7-16	E7-16	K7-16	T7-16	J7-16	1.000
A7-18	AD7-18	B7-18	D7-18	DD7-18	E7-18	K7-18	T7-18	J7-18	1.125
A7-20	AD7-20	B7-20	D7-20	DD7-20	E7-20	K7-20	T7-20	J7-20	1.250
A7-22	AD7-22	B7-22	D7-22	DD7-22	E7-22	K7-22	T7-22	J7-22	1.375
A7-24	AD7-24	B7-24	D7-24	DD7-24	E7-24	K7-24	T7-24	J7-24	1.500
A7-28	AD7-28	B7-28	D7-28	DD7-28	E7-28	K7-28	T7-28	J7-28	1.750
A7-32	AD7-32	B7-32	D7-32	DD7-32	E7-32	K7-32	T7-32	J7-32	2.000
A7-40	AD7-40	B7-40	D7-40	DD7-40	E7-40	K7-40	T7-40	J7-40	2.500
A7-48	AD7-48	B7-48	D7-48	DD7-48	E7-48	K7-48	T7-48	J7-48	3.000
A7-56	AD7-56	B7-56	D7-56	DD7-56	E7-56	K7-56	T7-56	J7-56	3.500

REVISION

3

NASM20470

SHEET 7

USE OF OR RELIANCE UPON THIS DOCUMENT OR ANY NATIONAL AEROSPACE STANDARD IS ENTIRELY VOLUNTARY. AIA DOES NOT QUALIFY SUPPLIERS OR CERTIFY CONFORMANCE OF ITEMS PRODUCED UNDER NATIONAL AEROSPACE STANDARDS. AIA MAKES NO REPRESENTATION OR CLAIM RESPECTING (1) THE SUITABILITY OF ITEMS FOR ANY PARTICULAR USE OR (2) THE ACCURACY OF THE INFORMATION CONTAINED HEREIN. THIS DOCUMENT IS THE PROPERTY OF AIA AND IS LOANED TO YOU. IT IS TO BE RETURNED TO AIA UPON REQUEST. No reproduction or networking permitted without license from Accuris

Not for Resale, 10/10/2025 23:47:03 MDT

AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA, INC.
1000 WILSON BLVD.
ARLINGTON, VA 22209

AIRBUS | NormMaster

THIS DRAWING SUPERSEDES ALL ANTECEDENT STANDARD DRAWINGS FOR THE SAME PRODUCT AND SHALL BECOME EFFECTIVE NO LATER THAN SIX MONTHS FROM THE LAST REVISION DATE.

FORM 21-01



NATIONAL AEROSPACE STANDARD

© COPYRIGHT 2025 AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA, INC. ALL RIGHTS RESERVED



TABLE II – DASH NUMBER LENGTH DIMENSIONS (CONTINUED)

ALUMINUM 1100-F	ALUMINUM 2117-T4	ALUMINUM 5056-H32	ALUMINUM 2017-T4 INACTIVE	ALUMINUM 2024-T4 INACTIVE	ALUMINUM 7050-T73	ALUMINUM 2219-T81 INACTIVE	TITANIUM 45Cb	ALUMINUM 2219-T62	LENGTH
DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	
Ø1/4 NOM									
A8-6	AD8-6	B8-6	D8-6	DD8-6	E8-6	K8-6	T8-6	J8-6	.375
A8-7	AD8-7	B8-7	D8-7	DD8-7	E8-7	K8-7	T8-7	J8-7	.438
A8-8	AD8-8	B8-8	D8-8	DD8-8	E8-8	K8-8	T8-8	J8-8	.500
A8-9	AD8-9	B8-9	D8-9	DD8-9	E8-9	K8-9	T8-9	J8-9	.562
A8-10	AD8-10	B8-10	D8-10	DD8-10	E8-10	K8-10	T8-10	J8-10	.625
A8-12	AD8-12	B8-12	D8-12	DDB-12	E8-12	K8-12	T8-12	J8-12	.750
A8-14	AD8-14	B8-14	D8-14	DD8-14	E8-14	K8-14	T8-14	J8-14	.875
A8-16	AD8-16	B8-16	D8-16	DD8-16	E8-16	K8-16	T8-16	J8-16	1.000
A8-18	AD8-18	B8-18	D8-18	DD8-18	E8-18	K8-18	T8-18	J8-18	1.125
A8-20	AD8-20	B8-20	D8-20	DD8-20	E8-20	K8-20	T8-20	J8-20	1.250
A8-22	AD8-22	B8-22	D8-22	DD8-22	E8-22	K8-22	T8-22	J8-22	1.375
A8-24	AD8-24	B8-24	D8-24	DD8-24	E8-24	K8-24	T8-24	J8-24	1.500
A8-28	AD8-28	B8-28	D8-28	DD8-28	E8-28	K8-28	T8-28	J8-28	1.750
A8-32	AD8-32	B8-32	D8-32	DD8-32	E8-32	K8-32	T8-32	J8-32	2.000
A8-40	AD8-40	B8-40	D8-40	DD8-40	E8-40	K8-40	T8-40	J8-40	2.500
A8-48	AD8-48	B8-48	D8-48	DD8-48	E8-48	K8-48	T8-48	J8-48	3.000
A8-56	AD8-56	B8-56	D8-56	DD8-56	E8-56	K8-56	T8-56	J8-56	3.500

ALUMINUM 1100-F	ALUMINUM 2117-T4	ALUMINUM 5056-H32	ALUMINUM 2017-T4 INACTIVE	ALUMINUM 2024-T4 INACTIVE	ALUMINUM 7050-T73	ALUMINUM 2219-T81 INACTIVE	TITANIUM 45Cb	ALUMINUM 2219-T62	LENGTH
DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	
Ø5/16 NOM									
A10-8	AD10-8	B10-8	D10-8	DD10-8	E10-8	K10-8	T10-8	J10-8	.500
A10-9	AD10-9	B10-9	D10-9	DD10-9	E10-9	K10-9	T10-9	J10-9	.562
A10-10	AD10-10	B10-10	D10-10	DD10-10	E10-10	K10-10	T10-10	J10-10	.625
A10-12	AD10-12	B10-12	D10-12	DD10-12	E10-12	K10-12	T10-12	J10-12	.750
A10-14	AD10-14	B10-14	D10-14	DD10-14	E10-14	K10-14	T10-14	J10-14	.875
A10-16	AD10-16	B10-16	D10-16	DD10-16	E10-16	K10-16	T10-16	J10-16	1.000
A10-18	AD10-18	B10-18	D10-18	DD10-18	E10-18	K10-18	T10-18	J10-18	1.125
A10-20	AD10-20	B10-20	D10-20	DD10-20	E10-20	K10-20	T10-20	J10-20	1.250
A10-22	AD10-22	B10-22	D10-22	DD10-22	E10-22	K10-22	T10-22	J10-22	1.375
A10-24	AD10-24	B10-24	D10-24	DD10-24	E10-24	K10-24	T10-24	J10-24	1.500
A10-28	AD10-28	B10-28	D10-28	DD10-28	E10-28	K10-28	T10-28	J10-28	1.750
A10-32	AD10-32	B10-32	D10-32	DD10-32	E10-32	K10-32	T10-32	J10-32	2.000
A10-40	AD10-40	B10-40	D10-40	DD10-40	E10-40	K10-40	T10-40	J10-40	2.500
A10-48	AD10-48	B10-48	D10-48	DD10-48	E10-48	K10-48	T10-48	J10-48	3.000
A10-56	AD10-56	B10-56	D10-56	DD10-56	E10-56	K10-56	T10-56	J10-56	3.500
A10-64	AD10-64	B10-64	D10-64	DD10-64	E10-64	K10-64	T10-64	J10-64	4.000

REVISION

3

NASM20470

SHEET 8

USE OF OR RELIANCE UPON THIS DOCUMENT OR ANY NATIONAL AEROSPACE STANDARD IS ENTIRELY VOLUNTARY. AIA DOES NOT QUALIFY SUPPLIERS OR CERTIFY CONFORMANCE OF ITEMS PRODUCED UNDER NATIONAL AEROSPACE STANDARDS. AIA MAKES NO REPRESENTATION OR CLAIM RESPECTING (1) THE SUITABILITY OF ITEMS FOR ANY PARTICULAR APPLICATION OR (2) THE ACCURACY OF THE INFORMATION CONTAINED HEREIN. AIA ASSUMES NO LIABILITY FOR ANY DAMAGE, LOSS, OR INJURY, INCLUDING REASONABLE ATTORNEY'S FEES, ARISING OUT OF OR FROM THE USE OF THIS DOCUMENT OR ANY NATIONAL AEROSPACE STANDARD. AIA DOES NOT WARRANT THE ACCURACY OF THE INFORMATION CONTAINED HEREIN. AIA ASSUMES NO LIABILITY FOR ANY DAMAGE, LOSS, OR INJURY, INCLUDING REASONABLE ATTORNEY'S FEES, ARISING OUT OF OR FROM THE USE OF THIS DOCUMENT OR ANY NATIONAL AEROSPACE STANDARD.



NATIONAL AEROSPACE STANDARD

© COPYRIGHT 2025 AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA, INC. ALL RIGHTS RESERVED



TABLE II – DASH NUMBER LENGTH DIMENSIONS (CONTINUED)

ALUMINUM 1100-F	ALUMINUM 2117-T4	ALUMINUM 5056-H32	ALUMINUM 2017-T4 INACTIVE	ALUMINUM 2024-T4 INACTIVE	ALUMINUM 7050-T73	ALUMINUM 2219-T81 INACTIVE	TITANIUM 45Cb	ALUMINUM 2219-T62	LENGTH
DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	
Ø3/8 NOM									
A12-10	AD12-10	B12-10	D12-10	DD12-10	E12-10	K12-10	T12-10	J12-10	.625
A12-12	AD12-12	B12-12	D12-12	DD12-12	E12-12	K12-12	T12-12	J12-12	.750
A12-14	AD12-14	B12-14	D12-14	DD12-14	E12-14	K12-14	T12-14	J12-14	.875
A12-16	AD12-16	B12-16	D12-16	DD12-16	E12-16	K12-16	T12-16	J12-16	1.000
A12-18	AD12-18	B12-18	D12-18	DD12-18	E12-18	K12-18	T12-18	J12-18	1.125
A12-20	AD12-20	B12-20	D12-20	DD12-20	E12-20	K12-20	T12-20	J12-20	1.250
A12-22	AD12-22	B12-22	D12-22	DD12-22	E12-22	K12-22	T12-22	J12-22	1.375
A12-24	AD12-24	B12-24	D12-24	DD12-24	E12-24	K12-24	T12-24	J12-24	1.500
A12-28	AD12-28	B12-28	D12-28	DD12-28	E12-28	K12-28	T12-28	J12-28	1.750
A12-32	AD12-32	B12-32	D12-32	DD12-32	E12-32	K12-32	T12-32	J12-32	2.000
A12-40	AD12-40	B12-40	D12-40	DD12-40	E12-40	K12-40	T12-40	J12-40	2.500
A12-48	AD12-48	B12-48	D12-48	DD12-48	E12-48	K12-48	T12-48	J12-48	3.000
A12-56	AD12-56	B12-56	D12-56	DD12-56	E12-56	K12-56	T12-56	J12-56	3.500
A12-64	AD12-64	B12-64	D12-64	DD12-64	E12-64	K12-64	T12-64	J12-64	4.000

REVISION
3

NASM20470
SHEET 9

USE OF OR RELIANCE UPON THIS DOCUMENT OR ANY NATIONAL AEROSPACE STANDARD IS ENTIRELY VOLUNTARY. AIA DOES NOT QUALIFY SUPPLIERS OR CERTIFY CONFORMANCE OF ITEMS PRODUCED UNDER NATIONAL AEROSPACE STANDARDS. AIA MAKES NO REPRESENTATION OR CLAIM RESPECTING (1) THE SUITABILITY OF ITEMS FOR ANY PARTICULAR APPLICATION OR (2) THE ACCURACY OF THE INFORMATION CONTAINED HEREIN. AIA IS NOT RESPONSIBLE FOR ANY ERRORS OR OMISSIONS. AIA IS NOT RESPONSIBLE FOR ANY REPRODUCTION OR NETWORKING PERMITTED WITHOUT LICENSE FROM ACCURIS.

AIRBUS | NormMaster

THIS DRAWING SUPERSEDES ALL ANTECEDENT STANDARD DRAWINGS FOR THE SAME PRODUCT AND SHALL BECOME EFFECTIVE NO LATER THAN SIX MONTHS FROM THE LAST REVISION DATE.

FORM 21-01